

Math 81: Abstract Algebra

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Problem 1. Let $f(x) \in \mathbb{R}[x]$ be a monic polynomial of degree $n \geq 1$ with discriminant Δ .

- (a) Assume that $f(x)$ has no repeated roots and let r_2 be the number of pairs of complex conjugate (nonreal) roots. Prove that the sign of Δ is $(-1)^{r_2}$.
- (b) Let $f(x) \in \mathbb{R}[x]$ be a cubic polynomial. Prove that $\Delta \geq 0$ if and only if the roots of $f(x)$ are all real.

Solution.

- (a) Let $f(x) \in \mathbb{R}[x]$ be a separable monic polynomial of degree $n \geq 1$, and let r_2 be the number of pairs of complex conjugate roots. Note that since $f(x)$ has real coefficients, complex roots come in pairs. The discriminant of $f(x)$ is defined as

$$\Delta = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

where α s are the roots of the polynomial. Each term in the product falls into one of these cases:

- (a) Both α_i and α_j are real:

Then term $(\alpha_i - \alpha_j)^2$ is a positive real number.

- (b) Both α_i and α_j are complex:

If $\alpha_j = \overline{\alpha_i}$, then

$$(\alpha_i - \alpha_j)^2 = (2i\text{Im}(\alpha_i))^2 = -4\text{Im}(\alpha_i)^2 < 0$$

where $\text{Im}(\alpha_i)$ is the coefficient of the imaginary part of α_i .

If α_i and α_j are not conjugates, then their conjugates $\overline{\alpha_i}$ and $\overline{\alpha_j}$ also appear among the roots, and the product

$$(\alpha_i - \alpha_j)^2(\overline{\alpha_i} - \overline{\alpha_j})^2 = ((\alpha_i - \alpha_j)(\overline{\alpha_i} - \overline{\alpha_j}))^2 > 0$$

is the square of a real number, hence positive.

(c) (WLOG) α_i is real and α_j is complex.

Then $\overline{\alpha_j}$ is also a root, and the product

$$(\alpha_i - \alpha_j)^2(\alpha_i - \overline{\alpha_j})^2 = (\alpha_i^2 - (\alpha_j + \overline{\alpha_j})\alpha_i + \alpha_j\overline{\alpha_j})^2 > 0$$

since both $(\alpha_j + \overline{\alpha_j})$, and $\alpha_j\overline{\alpha_j}$ are real numbers.

That is the only term in the product that are not positive are the ones formed by complex conjugates (each pair contributing -1 to Δ), and since there are r_2 pairs, the sign of the discriminant is $(-1)^{r_2}$.

(b) Let $f(x) \in \mathbb{R}[x]$ be a cubic polynomial. Note that since complex roots occur in pairs, $f(x)$ either has all real roots, or one pair of complex roots and 1 real root.

($\implies$) Assume that $\Delta(f) \geq 0$.

Then from part (a) we know that, $\text{sgn}(\Delta(f)) = (-1)^{r_2}$ where r_2 is the number of pairs of complex roots.

For a cubic polynomial, r_2 is either 0 or 1. If $r_2 = 1$, then

$$\text{sgn}(\Delta(f)) = (-1)^1 = -1,$$

so $\Delta(f) < 0$. (Note that $\Delta(f) \neq 0$ because the roots would all be distinct.) This contradicts the assumption that $\Delta(f) \geq 0$.

Hence $r_2 = 0$, and therefore all roots of $f(x)$ are real.

($\impliedby$) Assume all roots of $f(x)$ are real. Let the be $\alpha_1, \alpha_2, \alpha_3$.

Then,

$$\Delta(f) = \prod_{1 \leq i < j \leq 3} (\alpha_i - \alpha_j)^2$$

where each $\alpha_i - \alpha_j$ is a real number. Squares of real numbers are always positive. That is, $\Delta(f) \geq 0$.

Problem 2. Let $\gamma = \sqrt{2 + \sqrt{2}} \in \mathbb{R}$.

- (a) Show that $\mathbb{Q}(\gamma)/\mathbb{Q}$ is Galois with cyclic Galois group.
- (b) Show that $\mathbb{Q}(\gamma, i) = \mathbb{Q}(\zeta_{16})$ and compute how each element of the Galois group $\text{Gal}(\mathbb{Q}(\zeta_{16})/\mathbb{Q})$ acts on γ and i .

Solution.

- (a) Let $f(x) = x - \gamma \in \mathbb{Q}(\gamma)[x]$

$$\begin{aligned} x - \gamma &= 0 \\ x^2 &= 2 + \sqrt{2} \\ x^4 &= 4 + 2 + 4\sqrt{2} = 4x^2 - 2 \\ g(x) &= x^4 - 4x^2 + 2 \end{aligned}$$

Note that $g(x) \in \mathbb{Q}[x]$, and has γ as a root. Also note that since $g(x)$ is irreducible by Eisenstein, it is the minimal polynomial of γ over $\mathbb{Q}$.

We know that $g(x)$ is separable because $g(x)$ and $g'(x) = 4x(x^2 - 2)$ have no common factors. That is $\gcd(g(x), g'(x)) = 1$. Now, let K be the splitting field of $g(x)$.

Consider $x^2 = y$, then $g(y) = y^2 + 4y + 2$ with roots $2 + \sqrt{2}$ and $2 - \sqrt{2}$. That is,

$$x^2 = 2 + \sqrt{2}, \quad x^2 = 2 - \sqrt{2}$$

giving us,

$$x = \pm\sqrt{2 + \sqrt{2}}, \quad x = \pm\sqrt{2 - \sqrt{2}}$$

now note, $\gamma^2 - 2 = \sqrt{2} \in \mathbb{Q}(\gamma)$, and $\sqrt{2 - \sqrt{2}} = \sqrt{2}/\gamma \in \mathbb{Q}(\gamma)$. That is $K = \mathbb{Q}(\gamma)$. Since $\mathbb{Q}(\gamma)$ is the splitting field of a separable polynomial, it is Galois.

Now note, $|\text{Gal}(\mathbb{Q}(\gamma)/\mathbb{Q})| = [\mathbb{Q}(\gamma) : \mathbb{Q}] = 4$. So the group is either C_4 or V_4 , that is showing that there exists an element of order > 2 suffices to show that the group is C_4 . Consider the automorphism $\sigma \in \text{Gal}(\mathbb{Q}(\gamma)/\mathbb{Q})$:

$$\sigma : \sqrt{2 + \sqrt{2}} \mapsto \sqrt{2 - \sqrt{2}}$$

And $\sigma((\sqrt{2 + \sqrt{2}})^2) = (\sqrt{2 - \sqrt{2}})^2$, that is $\sqrt{2} \mapsto -\sqrt{2}$, and $\sigma(\sqrt{2 - \sqrt{2}}) = -\sqrt{2}/(\sqrt{2 + \sqrt{2}})$

$$\begin{aligned} \sigma(\sqrt{2 + \sqrt{2}}) &= \sqrt{2 - \sqrt{2}} \\ \sigma^2(\sqrt{2 + \sqrt{2}}) &= \sigma(\sqrt{2 - \sqrt{2}}) = -\sqrt{2}/(\sqrt{2 + \sqrt{2}}) = -(\sqrt{2 + \sqrt{2}}) \end{aligned}$$

That is $|\sigma| \neq 2$ and since it is not the identity, it must have order > 2 , i.e. the Galois group of $\mathbb{Q}(\gamma)$ is $\mathbb{C}_4$.

(b) $\mathbb{Q}(\gamma, \mathbf{i}) \subseteq \mathbb{Q}(\zeta_{16})$.

$$\zeta_{16}^4 = \mathbf{i}, \quad \mathbf{i} \in \mathbb{Q}(\zeta_{16})$$

And,

$$\begin{aligned} \zeta_{16} &= \cos\left(\frac{2\pi}{16}\right) + \mathbf{i} \sin\left(\frac{2\pi}{16}\right) = \cos\left(\frac{\pi}{8}\right) + \mathbf{i} \sin\left(\frac{\pi}{8}\right) \\ \zeta_{16}^{15} &= \cos\left(2\pi - \left(\frac{\pi}{16}\right)\right) + \mathbf{i} \sin\left(2\pi - \left(\frac{\pi}{16}\right)\right) = \cos\left(\frac{\pi}{8}\right) - \mathbf{i} \sin\left(\frac{\pi}{8}\right) \end{aligned}$$

But

$$\begin{aligned} \cos\left(\frac{\pi}{8}\right) &= \sqrt{\frac{1 + \cos\left(\frac{\pi}{4}\right)}{2}} \\ &= \sqrt{\frac{\frac{2+\sqrt{2}}{2}}{2}} = \sqrt{\frac{2+\sqrt{2}}{4}} = \frac{\sqrt{2+\sqrt{2}}}{2} = \frac{\gamma}{2} \end{aligned}$$

And

$$\begin{aligned} \sin\left(\frac{\pi}{8}\right) &= \sqrt{\frac{1 - \cos\left(\frac{\pi}{4}\right)}{2}} \\ &= \sqrt{\frac{\frac{2-\sqrt{2}}{2}}{2}} = \sqrt{\frac{2-\sqrt{2}}{4}} = \frac{\sqrt{2-\sqrt{2}}}{2} \end{aligned}$$

Which gives us,

$$\zeta_{16} + \zeta_{16}^{15} = 2 \cdot \frac{\sqrt{2+\sqrt{2}}}{2} = \gamma$$

Hence $\gamma, \mathbf{i} \in \mathbb{Q}(\zeta_{16})$

$\mathbb{Q}(\gamma, \mathbf{i}) \supseteq \mathbb{Q}(\zeta_{16})$.

$$\begin{aligned} \zeta_{16} &= \cos\left(\frac{2\pi}{16}\right) + \mathbf{i} \sin\left(\frac{2\pi}{16}\right) \\ &= \cos\left(\frac{\pi}{8}\right) + \mathbf{i} \sin\left(\frac{\pi}{8}\right) \end{aligned}$$

But

$$\begin{aligned} \cos\left(\frac{\pi}{8}\right) &= \sqrt{\frac{1 + \cos\left(\frac{\pi}{4}\right)}{2}} \\ &= \sqrt{\frac{\frac{2+\sqrt{2}}{2}}{2}} = \sqrt{\frac{2+\sqrt{2}}{4}} = \frac{\sqrt{2+\sqrt{2}}}{2} \end{aligned}$$

And

$$\begin{aligned}\sin\left(\frac{\pi}{8}\right) &= \sqrt{\frac{1 - \cos\left(\frac{\pi}{4}\right)}{2}} \\ &= \sqrt{\frac{\frac{2-\sqrt{2}}{2}}{2}} = \sqrt{\frac{2-\sqrt{2}}{4}} = \frac{\sqrt{2-\sqrt{2}}}{2}\end{aligned}$$

Which gives us,

$$\begin{aligned}\zeta_{16} &= \cos\left(\frac{\pi}{8}\right) + i \sin\left(\frac{\pi}{8}\right) \\ &= \frac{\sqrt{2+\sqrt{2}}}{2} + i \frac{\sqrt{2-\sqrt{2}}}{2}\end{aligned}$$

Note from part(a), we know that $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\gamma)$ so $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\gamma, i)$ and hence $\zeta_{16} \in \mathbb{Q}(\gamma, i)$.

Note that $\text{Gal}(\mathbb{Q}(\zeta_{16})/\mathbb{Q}) \cong (\mathbb{Z}/16\mathbb{Z})^\times$ (Theorem 26). Where,

$$\begin{aligned}\varphi : (\mathbb{Z}/16\mathbb{Z})^\times &\rightarrow \text{Gal}(\mathbb{Q}(\zeta_{16})/\mathbb{Q}) \\ \mathfrak{a} \pmod{16} &\mapsto \sigma_{\mathfrak{a}}\end{aligned}$$

where $\sigma_{\mathfrak{a}}$ is the automorphism defined by $\sigma_{\mathfrak{a}}(\zeta_{16}) = \zeta_{16}^{\mathfrak{a}}$. That is the elements of the Galois group take primitive n -th roots to primitive n -th roots.

Action on i :

$$\sigma_{\mathfrak{a}}(i) = \sigma_{\mathfrak{a}}(\zeta_{16}^4) = \zeta_{16}^{4\mathfrak{a}}$$

where $4\mathfrak{a}$ is computed mod 16.

That is, if $\mathfrak{a} \equiv 1, 5, 9, 13 \pmod{16}$ then $4\mathfrak{a} \equiv 4 \pmod{16}$ and $\zeta_{16}^4 = i$. Else if $\mathfrak{a} \equiv 3, 7, 11, 15 \pmod{16}$ then $4\mathfrak{a} \equiv 12 \pmod{16}$ and $\zeta_{16}^{12} = -i$.

Action on γ :

$$\sigma_{\mathfrak{a}}(\gamma) = \sigma_{\mathfrak{a}}(\zeta_{16} + \zeta_{16}^{-1}) = \zeta_{16}^{\mathfrak{a}} + \zeta_{16}^{-\mathfrak{a}} = 2 \cos\left(\frac{2\pi\mathfrak{a}}{16}\right)$$

where $\mathfrak{a}$ is computed mod 16.

In particular,

$$\begin{aligned}\mathfrak{a} \equiv 1, 15 \pmod{16} &\Rightarrow \sigma_{\mathfrak{a}}(\gamma) = \zeta_{16} + \zeta_{16}^{-1}, \\ \mathfrak{a} \equiv 3, 13 \pmod{16} &\Rightarrow \sigma_{\mathfrak{a}}(\gamma) = \zeta_{16}^3 + \zeta_{16}^{-3}, \\ \mathfrak{a} \equiv 5, 11 \pmod{16} &\Rightarrow \sigma_{\mathfrak{a}}(\gamma) = \zeta_{16}^5 + \zeta_{16}^{-5}, \\ \mathfrak{a} \equiv 7, 9 \pmod{16} &\Rightarrow \sigma_{\mathfrak{a}}(\gamma) = \zeta_{16}^7 + \zeta_{16}^{-7}.\end{aligned}$$

Problem 3. In this problem, you'll see a polynomial whose roots are solvable by radicals but whose splitting field is not itself a radical extension. Let $K = \mathbb{Q}(\zeta_7)$ and $L = \mathbb{Q}(\zeta_7 + \overline{\zeta_7})$.

- (a) Prove that $K/\mathbb{Q}$ is a radical extension.
- (b) Prove that $L/\mathbb{Q}$ is not a radical extension.

Warning: A simple extension $F(\alpha)$ can be radical even if α is not an n th root of anything in F (try to think of an example).

- (c) Find the minimal polynomial of $\zeta_7 + \overline{\zeta_7}$. Hint. $\zeta_7^4 = \overline{\zeta_7}^3$.
- (d) Write down a polynomial of degree 3 over $\mathbb{Q}$ (that is solvable by radicals but) whose splitting field is not a radical extension of $\mathbb{Q}$.

Solution.

- (a) Let $K = \mathbb{Q}(\zeta_7)$. Now note $\zeta_7^7 = 1 \in \mathbb{Q}$. Since the element ζ_7 is a 7th root of an element of $\mathbb{Q}$ it is obtained from $\mathbb{Q}$ by adjoining a radical, and hence is a radical extension.
- (b) Since ζ_7 is a primitive root of unity, we know,

$$[K/\mathbb{Q}] = \varphi(7) = 6$$

and,

$$\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times$$

Now, since $\mathbb{F} \subseteq L \subseteq K$ and $\zeta_7 + \overline{\zeta_7} = 2 \cos(2\pi/7) \in \mathbb{R}$, $[K : L] = 2$ and by Tower Law, $[L : \mathbb{F}]$ must be 3.

For the sake of contradiction, assume L is radical. Since 3 is prime, by Tower Law there are no non-trivial extensions then $\zeta_7 + \overline{\zeta_7} = \sqrt[3]{a}$, for some $a \in \mathbb{Q}$. Moreover, $L/\mathbb{Q}$ is Galois (it is the fixed subfield of an abelian Galois group). But if this extension is Galois, then L must have all roots of $x^3 - a$, which are $\sqrt[3]{a}$, $\zeta_3 \sqrt[3]{a}$, $\zeta_3^2 \sqrt[3]{a}$. But $L \subset \mathbb{R}$, and hence a contradiction.

That is, $L/\mathbb{Q}$ is not a radical extensions.

- (c) Note that $\overline{\zeta_7} = \zeta_7^6$, and $\Phi_7(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$, that is

$$\zeta_7^6 + \zeta_7^5 + \zeta_7^4 + \zeta_7^3 + \zeta_7^2 + \zeta_7 + 1 = 0$$

$$(\zeta_7^6 + \zeta_7) + (\zeta_7^4 + \zeta_7^3) + (\zeta_7^2 + \zeta_7^5) + 1 = 0$$

Also,

$$(\zeta_7^6 + \zeta_7)^2 = \zeta_7^5 + \zeta_7^2 + 2, \quad (\zeta_7^6 + \zeta_7)^3 = \zeta_7^4 + \zeta_7^3 + 3(\zeta_7^6 + \zeta_7)$$

Substituting back, $\mathfrak{x} = \zeta_7^6 + \zeta_7$,

$$\mathfrak{x} + \mathfrak{x}^3 - 3\mathfrak{x} + \mathfrak{x}^2 - 2 + 1 = \mathfrak{x}^3 + \mathfrak{x}^2 - 2\mathfrak{x} - 1 = 0$$

The minimal polynomial is $\mathfrak{x}^3 + \mathfrak{x}^2 - 2\mathfrak{x} - 1 = 0$.

(d) $f(\mathfrak{x}) = \mathfrak{x}^3 + \mathfrak{x}^2 - 2\mathfrak{x} - 1$

Problem 4. Let $f(x) \in \mathbb{Q}[x]$ be an irreducible quartic polynomial and $K = \mathbb{Q}(\alpha)$ for a root α of $f(x)$. Let $G \subset S_4$ be the Galois group of the splitting field of $f(x)$ over $\mathbb{Q}$.

- (a) Prove that $K/\mathbb{Q}$ has no nontrivial intermediate subfields if and only if $G = A_4$ or $G = S_4$.
- (b) Prove that α is constructible if and only if the normal closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ has Galois group C_4 , V_4 , or D_8 .
- (c) Find a nonconstructible $\alpha \in \mathbb{C}$ with $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$.

Solution.

- (a) ($\implies$) Assume $K/\mathbb{Q}$ has no non-trivial intermediate subfields.

L be the splitting field of $f(x)$ over $\mathbb{Q}$. That is $\mathbb{Q} \subset K \subseteq L$. And let $G = \text{Gal}(L/\mathbb{Q}) \subseteq S_4$. Since $f(x)$ is irreducible, we know that $[K : \mathbb{Q}] = 4$, and also that G is transitive.

Let $H \subset G$ be the group that fixes K in L . Then by Galois correspondence, if $K/\mathbb{Q}$ has no non-trivial subfields, there are no groups between H and G . From FTGT, the following holds

$$[K : \mathbb{Q}] = [G : H] = 4$$

Now consider the transitive subgroups of S_4 .

- (i) C_4 - A subgroup of index 4 would have order 1. There is a subgroup of order 2 between 1 and C_4 and hence there exist a corresponding nontrivial intermediate subfield.
- (ii) V_4 - A subgroup of index 4 would have order 1. There are 3 subgroups of order 2 between 1 and V_4 and hence there exist corresponding nontrivial intermediate subfields.
- (iii) D_8 - A subgroup of index 4 would have order 2. There are several subgroups of order 4 between 1 and D_4 and hence there exist corresponding nontrivial intermediate subfields.
- (iv) A_4 - A subgroup of index 4 would have order 3. There is no subgroup of size 6 in A_4 , so there are no subgroups and hence no intermediate fields.
- (v) S_4 - A subgroup of index 4 would have order 6. The stabilizer of a root is isomorphic to $S_3 \subset S_4$. And note that although there is a group of order 12 in S_4 , $S_3 \not\subseteq A_4$. Hence there are no intermediate fields.

Hence, the only possible transitive groups of S_4 that G can be is A_4 and S_4 .

($\Leftarrow$) Assume $G = A_4$ or $G = S_4$. Let L be the splitting field of $f(x)$ over $\mathbb{Q}$, and let $H = \text{Gal}(L/K)$. Then we have $F \subset K \subseteq L$, and $\{e\} \subset H \subset G$. Moreover, we know $[L : \mathbb{Q}] = |G|$, and $[K : \mathbb{Q}] = 4$, then by Tower Law $[L : K] = |G|/4$.

(i) $G = S_4$.

$[L : K] = 6 = |H|$, which means $|H| \cong S_3$. There is no subgroup of size 6 in A_4 . The stabilizer of a root is isomorphic to $S_3 \subset S_4$. Note that although there is a group of order 12 in S_4 , $S_3 \not\subseteq A_4$. Hence there are no intermediate fields.

(ii) $G = A_4$.

$[L : K] = 3$, that is $\text{Gal}(L/K) \cong C_3$. And there are no subgroups between C_3 and A_4 , i.e. there are no corresponding subfields between $K/\mathbb{Q}$.

Hence proved.

(b) ($\Rightarrow$) Assume α is constructible.

Then there exists a tower of quadratic extensions

$$\mathbb{Q} = F_0 \subset F_1 \subset \cdots \subset F_r$$

with $[F_{i+1} : F_i] = 2$ and $\alpha \in F_r$. That is $\mathbb{Q}(\alpha) \subseteq F_r$.

Since $K = \mathbb{Q}(\alpha)$ has $[K : \mathbb{Q}] = 4$, we have $K \subseteq F_r$. Let L be the normal closure of $K/\mathbb{Q}$ and let $G = \text{Gal}(L/\mathbb{Q})$. Since $L \subseteq F_r$ and $[F_r : \mathbb{Q}] = 2^r$, it follows that $|G| = [L : \mathbb{Q}]$ is a power of 2. Now consider the transitive subgroups of S_4 .

(i) $C_4 = \langle g \rangle$

$$1 \trianglelefteq \langle g^2 \rangle \trianglelefteq \langle g \rangle = C_4$$

And there exists a corresponding quadratic tower of extensions that reaches K .

(ii) $V_4 = \{1, a, b, ab\}$

$$1 \trianglelefteq \langle a \rangle \trianglelefteq \langle a, b \rangle = V_4$$

And there exists a corresponding quadratic tower of extensions that reaches K .

(iii) $D_8 = \langle s, r \mid e = s^2 = r^4, sr = r^{-1}s \rangle$

$$1 \trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r^2, s \rangle \triangleleft \langle r, s \rangle = D_8$$

And there exists a corresponding quadratic tower of extensions that reaches K .

(iv) A_4 - This group has order 12 which is not a power of 2, hence there doesn't exist such a sequence, and there does not exist a corresponding quadratic tower that reaches G , hence elements with this Galois groups are not constructible.

(v) S_4 - This group has order 24 which is not a power of 2, hence there doesn't exist such a sequence, and there does not exist a corresponding quadratic tower that reaches G , hence elements with this Galois groups are not constructible.

Hence $G \cong C_4, V_4$, or D_8 .

($\Leftarrow$) Assume that the normal closure L of $K/\mathbb{Q}$ has

$$G = \text{Gal}(L/\mathbb{Q}) \cong C_4, V_4, \text{ or } D_8.$$

Since these are all 2-groups, we can find a chain of subgroups

$$\{1\} = G_0 \subset G_1 \subset \cdots \subset G_n = G$$

with $[G_{i+1} : G_i] = 2$. By the Fundamental Theorem of Galois Theory, this corresponds to a tower of quadratic extensions

$$\mathbb{Q} = L^G \subset L^{G_{n-1}} \subset \cdots \subset L^{G_0} = L$$

where each extension has degree 2.

Since $K = \mathbb{Q}(\alpha)$ is an intermediate field of $L/\mathbb{Q}$, it lies in this tower. Hence $K/\mathbb{Q}$ is obtained by successive quadratic extensions, and therefore α is constructible.

Hence proved.

(c) Let $f(x) = x^4 - x - 1$, and α be a root to $f(x)$. Consider $f(x) \in \mathbb{F}_2[x]$. For $x = 0$:

$$f(0) = (0)^4 - 0 - 1 = 1$$

And, for $x = 1$:

$$f(1) = (1)^4 - 1 - 1 = -1$$

That is, $f(x)$ doesn't factor into a linear * cubic polynomials, or 2 linear * quadratic, or 4 linears. That only leaves us with it factoring into two quadratics, The only irreducible quadratic in $\mathbb{F}_2[x]$ is $g(x) = x^2 + x + 1$. Note that $g(x) \nmid f(x)$. Hence $f(x)$ is irreducible and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. To check separability we note that $f'(x) = 4x^3 - 1$, and $\gcd(f'(x), f(x)) = 1$, hence it is separable.

Let $g(x)$ be the cubic resolvent of $f(x)$. Then $g(x) = x^3 + 4x - 1$. This polynomial is also irreducible by RRT. And $\Delta(g) = \Delta(f) = -27(-1)^2 - 4(4)^3 = -283$, which is not a square in $\mathbb{Q}$. Then its Galois group is S_4 , and by part (b) we know that this is not constructible.

Problem 5. Let p be a prime number and S_p the symmetric group on p things.

- (a) Prove that an element of S_p has order p if and only if it is a p -cycle.
- (b) Prove that S_p is generated by any choice of element of order p and a transposition. Find a composite n and a choice of an n -cycle and a transposition that do not generate S_n .

Hint. For any n , DF 3.5 Exercise #4 says that S_n is generated by (12) and $(12 \cdots n)$. What is special about p being prime is that every element of order p in S_p is a p -cycle and every power of a p -cycle is a p -cycle (or the identity), which are facts you should verify. Hence up to conjugation and taking powers, you can assume that your transposition is (12) and that your p -cycle is $(12 \cdots)$. What then?

- (c) Let $F \subset \mathbb{R}$ be a subfield. Prove that if $f(x) \in F[x]$ is an irreducible polynomial of degree p having $p-2$ real roots, then the Galois group of $f(x)$ over F is isomorphic to S_p .
- (d) Let $F \subset \mathbb{R}$ be a subfield. Prove that if $f(x) \in F[x]$ is an irreducible cubic polynomial with $\Delta < 0$, then the Galois group of $f(x)$ over F is isomorphic to S_3 .
- (e) Prove that the Galois group of the polynomial $x^3 - x - 1$ over $\mathbb{Q}$ is isomorphic to S_3 .
- (f) Prove that the Galois group of the polynomial $x^5 - x^4 - x^2 - x + 1$ over $\mathbb{Q}$ is isomorphic to S_5 . Hint. You are allowed to use real analysis (e.g., the intermediate value theorem), but as a challenge, try to find a purely algebraic (possibly computer-aided) way.

Solution.

- (a) ($\implies$) Assume $\sigma \in S_p$ has order p . Recall that the order of a permutation is equal to the lcm of the lengths of its disjoint cycles. Since p is prime, the only such decomposition of σ would be into a 1-cycle (fixed points) and a p -cycle. Note that since there are p total objects there can be at most 1 p -cycle, i.e. σ cannot be decomposed into two or more disjoint p cycles.
- ($\impliedby$) The order of a cycle is equal to its length, i.e., a p -cycle has order p .
- (b) Let $\sigma = (a_1, \dots, a_p) \in S_p$ be a p cycle, and $\tau = (a_1, a_2)$ be a transposition. Recall from Math 71, that elements in S_n are conjugate if they have the same cycle type. That is p -cycles are conjugate in S_p and transpositions are also conjugate in S_p , or the conjugacy action acts transitively on these cycles. Then, upto conjugation we assume

$\sigma = (1, 2, \dots, p)$ and $\tau = (1, 2)$. Then by DF 3.5 Exercise #4, S_p is generated by $(1, 2, \dots, p)$ and $(1, 2)$.

- (c) Let $F \subset \mathbb{R}$, K be the splitting field of $f(x) \in F[x]$. Assume that $f(x)$ has $p - 2$ real roots, or $f(x)$ has two complex roots (which are complex conjugates). We know that the Galois group G of K is a subset of S_p .

Note that complex conjugation fixes F and the real roots, and hence is an automorphism. The order this automorphism is 2 since the complex conjugation of the complex conjugation is the original number. Since $f(x)$ is irreducible G acts transitively on the set of roots.

Claim: If $G \subseteq S_p$ is a transitive subgroup it contains a p cycle.

Proof: Assume G is a transitive subgroup. Then by orbit stabilizer

$$|G| = |\text{orb}(x)| \cdot |\text{stab}(x)| = p \cdot |\text{stab}(x)|$$

That is $p \mid |G|$. By Cauchy's Theorem, we know that every group with order divisible by p has an element of order p , and from part (a) we know that if any $x \in G \subseteq S_p$ has order p then it is a p -cycle. Hence proved.

That is, G has a 2-cycle and a p -cycle. From (b), we know that the subgroup of G generated by these cycles is S_p . Hence, $S_p \subseteq G \subseteq S_p$, so $G = S_p$.

- (d) Let $F \subset \mathbb{R}$, if $f(x) \in F[x]$ is an irreducible cubic with $\Delta < 0$ then from Problem 1 (b), we know that it has 2 complex roots. From part (c), this gives us that the Galois group of $f(x)$ over F is isomorphic to S_3 .

- (e) Let $f(x) = x^3 - x - 1$. Note that $f(x)$ is irreducible by RRT since $f(1) = -1$ and $f(-1) = -1$. Also, recall from pset 4 problem 3, that given a polynomial $g(x) = x^3 + px + q$, it's discriminant $\Delta(f)$ is $-4p^3 - 27q^2$.

That is, $\Delta(f) = -4(-1)^3 - 27(-1) = -23$. Since $\Delta(f) < 0$ and $f(x)$ irreducible, from part (d), we know that the Galois group of $f(x)$ is isomorphic to S_3 .

- (f) Let $f(x) = x^5 - x^4 - x^2 - x + 1$.

Consider $f(x) \in \mathbb{F}_2[x]$. Then,

$$f(0) = 0 - 0 - 0 - 0 + 1 = 1$$

and,

$$f(1) = 1 - 1 - 1 - 1 + 1 = -3 \equiv -1$$

since there are no roots in $\mathbb{F}_2[x]$, $f(x)$ can not factor as a linear * quartic, or a linear * 2 quadratics. However, we must still check that it doesn't factor as a degree 2 and a degree 3 polynomial. In $\mathbb{F}_2[x]$ the only irreducible quadratic is $g(x) = x^2 + x + 1$. Note $g(x) \nmid f(x)$ That is it is irreducible. Using computation, we can calculate that

$$\Delta(f) = -1 * 7 * 1217$$

Since, $\text{sgn}(\Delta(f)) = -1$, using Problem 1 part (a), we can say that this means that there is only 1 pair of complex conjugate roots (since f can have at most 2 pairs, and odd sign signifies odd number of pairs, we deduce that there is only 1 pair). That is there are $5 - 2$ real roots, and from part (c), we know that the Galois group of $f(x)$ over F is isomorphic to S_5 .